SME206
Signals and Systems

4. The Continuous-Time Fourier Transform

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Previously in Chapter 3…

- We developed a representation of **periodic signals** as **linear combinations of complex exponentials:**
  1) The signal must be **periodic**.

     In other words, there exists a $T > 0$ such that $x(t + T) = x(t)$
  2) The signal must **satisfy the Dirichlet conditions**.
- We also saw how this representation can be used in describing the effect of **LTI systems** on signals.

- In this chapter, we extend these concepts to signals that are **aperiodic**.

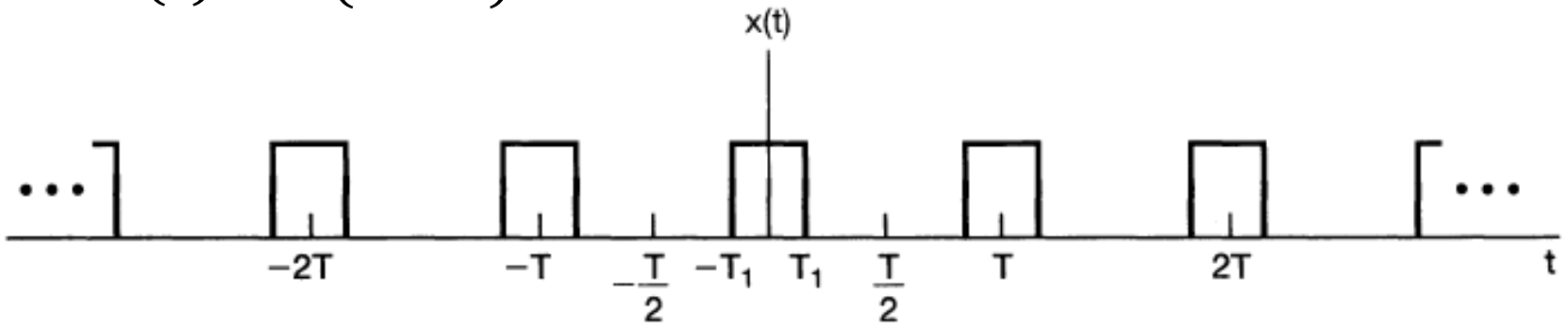
Developments from Fourier Series

- Consider the periodic signal

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

(Example 3.5)

where $x(t) = x(t+T)$.



- The FS coefficients of $x(t)$ are:

$$x(t) \overset{FS}{\longleftrightarrow} a_k = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} \qquad (k \neq 0)$$

Developments from Fourier Series

- Substituting $\omega = k\omega_0$, we have:

$$a_k = \left.\frac{2\sin(\omega T_1)}{\omega T}\right|_{\omega=k\omega_0}$$

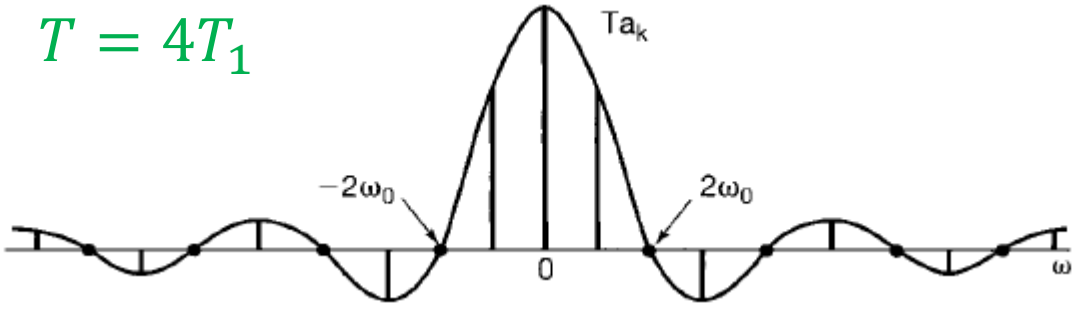
- Multiplying T on both sides:

$$Ta_k = \left.\frac{2\sin(\omega T_1)}{\omega}\right|_{\omega=k\omega_0}$$

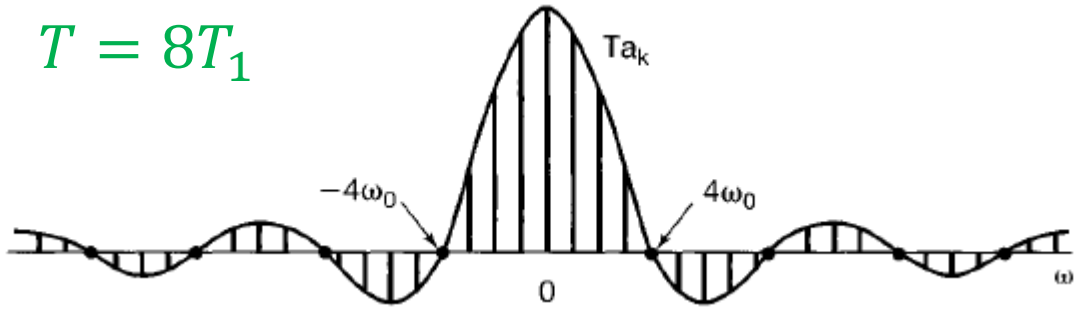
- Pictorially, with ω thought of as a continuous variable, the function $\frac{2sin(\omega T_1)}{\omega}$ represents the envelope of the series Ta_k

Developments from Fourier Series

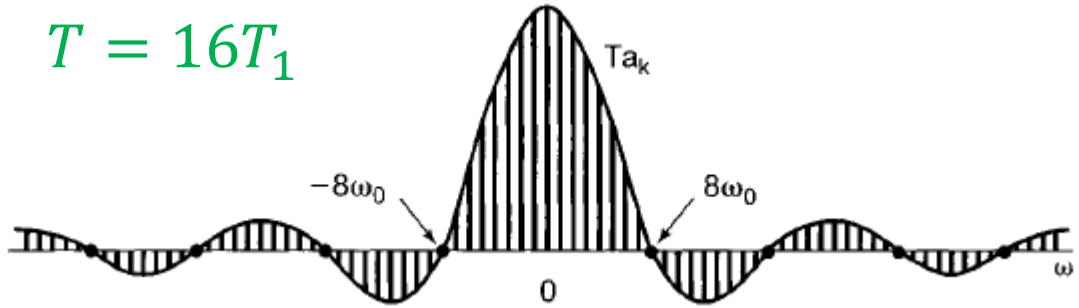
$T = 4T_1$



$T = 8T_1$



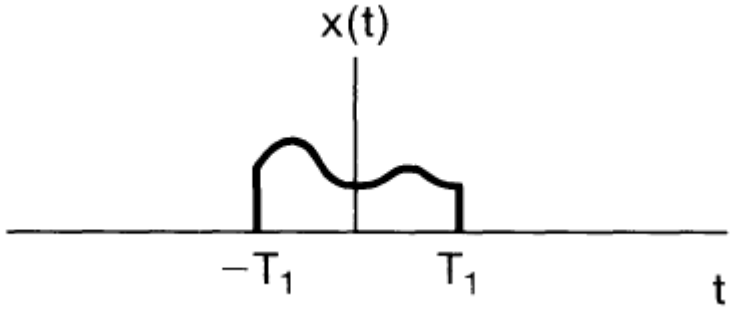
$T = 16T_1$



- **Envelope of Ta_k: $\frac{2sin(\omega T_1)}{\omega}$**
  - For fixed T_1, envelope is *independent* of T.
- **Coefficients a_k: equally placed samples of this envelope ($\omega = k\omega_0$).**
  - $T \uparrow \rightarrow \omega_0 \downarrow \rightarrow$ the envelop is sampled with an increasingly closer spacing
  - Ta_k approaches the envelope function as $T \rightarrow \infty$.

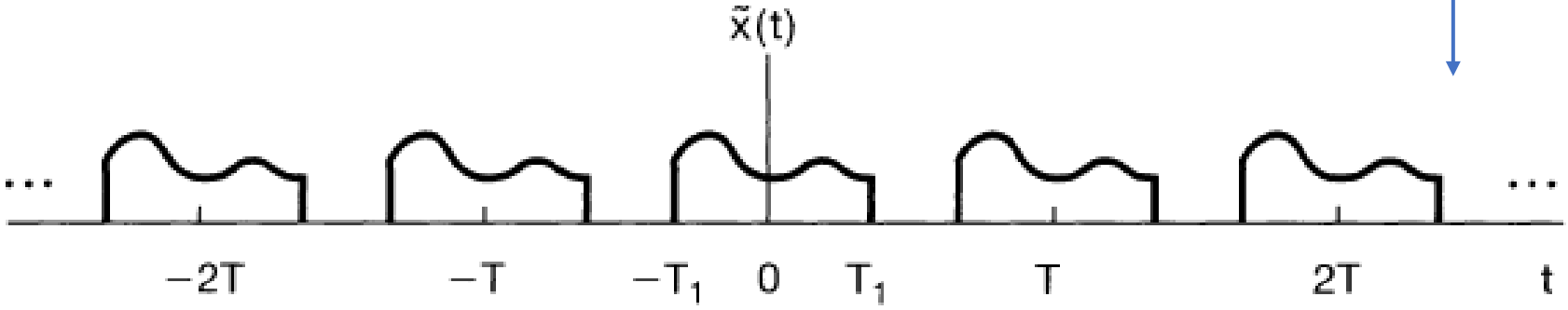
Fourier Transform: The Derivation

- Suppose an aperiodic signal $x(t)$:
  $x(t) = 0$ if $|t| > T_1$, for some number T_1



- From this aperiodic signal, let's construct a periodic signal $\tilde{x}(t)$ for which $x(t)$ is one period T

Period T



  - As $T \to \infty$, $\tilde{x}(t)$ is equal to $x(t)$ for any finite value of t.

Fourier Transform: The Derivation

- With $\tilde{x}(t)$ being periodic, we may express $\tilde{x}(t)$ using Fourier Series:

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}, \qquad where\ a_k = \frac{1}{T}\int_T \tilde{x}(t)e^{-jk\omega_0 t}dt$$

- With $\tilde{x}(t) = x(t)$ for $|t| < T/2$, and since $x(t) = 0$ outside this interval, we have:

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_{-\infty}^{+\infty} x(t)e^{-jk\omega_0 t}dt$$

- Therefore, defining the envelope $X(j\omega)$ of Ta_k as: $X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-jk\omega_0 t}dt$

We have, for the coefficients a_k: $a_k = \frac{1}{T}X(jk\omega_0)$

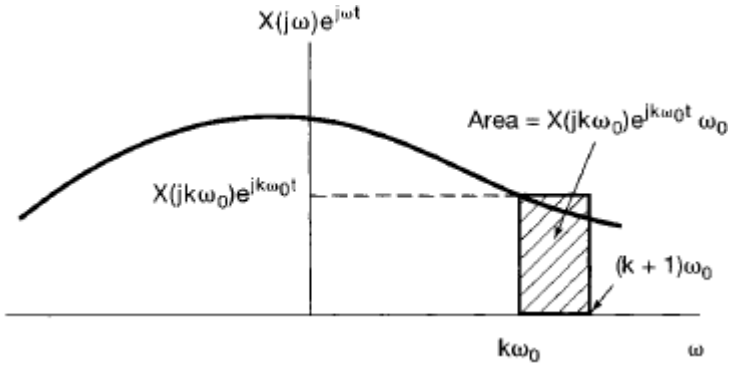
Fourier Transform: The Derivation

- With $a_k = \frac{1}{T}X(jk\omega_0)$, rewrite the Fourier Series for $\tilde{x}(t)$:

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t}$$

- Or equivalently, with $\omega_0 = 2\pi/T$,

$$\tilde{\boldsymbol{x}}(\boldsymbol{t}) = \frac{\mathbf{1}}{\mathbf{2\pi}} \sum_{\boldsymbol{k}=-\infty}^{+\infty} \boldsymbol{X(jk\omega_0) e^{jk\omega_0 t} \omega_0}$$



- As $T \to \infty$, $\tilde{x}(t)$ approaches $x(t)$, and $\omega_0 \to 0$:

$$\boldsymbol{x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega}, \qquad \boldsymbol{where\ X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt}$$

Fourier Transform

- **Summary**: Given a **continuous-time signal $x(t)$,**
  - The **Fourier Transform** of the signal is given by:

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt \qquad \text{(analysis equation)}$$

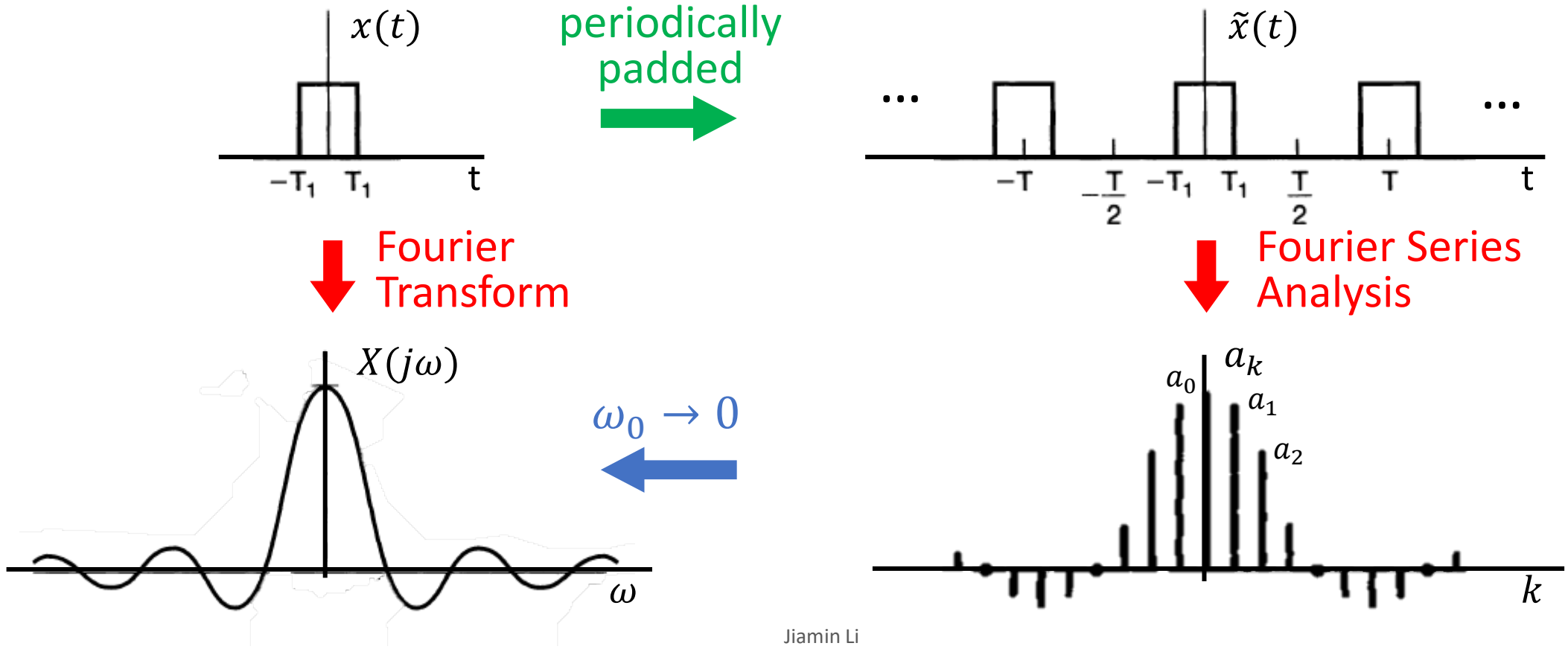
  - The **Inverse Fourier Transform** of the signal is given by:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega \qquad \text{(synthesis equation)}$$

- The **Fourier Transform $X(j\omega)$** is also called the **spectrum** of the signal, as it *describes* $x(t)$ as a linear combination (specifically, an integral) of sinusoidal signals at different frequencies.

Fourier Transform vs. Fourier Series: The Relationship

- **For an aperiodic signal: Only Fourier Transform is applicable**
  - Fourier Transform is *equivalent* to the limiting case (when $\omega_0 \to 0$) of applying Fourier Series analysis on the periodically padded version of the aperiodic signal.



Fourier Transform vs. Fourier Series: The Relationship

- **For a periodic signal: Both Fourier Transform & Fourier Series are applicable**
  - Consider a periodic signal $x(t)$ with period T (thus $\omega_0 = 2\pi/T$):
    - Apply FS analysis to $x(t)$:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

    - Apply Fourier Transform:

$$\boldsymbol{X(j\omega)} = \int_{-\infty}^{+\infty} \left(\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \right) e^{-j\omega t} dt = \sum_{k=-\infty}^{+\infty} a_k \left(\int_{-\infty}^{+\infty} e^{jk\omega_0 t} e^{-j\omega t} dt \right)$$

$$= \sum_{\boldsymbol{k=-\infty}}^{+\infty} \boldsymbol{a_k 2\pi \delta(\omega - k\omega_0)}$$

Fourier Transform vs. Fourier Series: The Relationship

- **Why is the below equality holds?**

$$\int_{-\infty}^{+\infty} e^{jk\omega_0 t} e^{-j\omega t} dt \Leftrightarrow 2\pi\delta(\omega - k\omega_0)$$

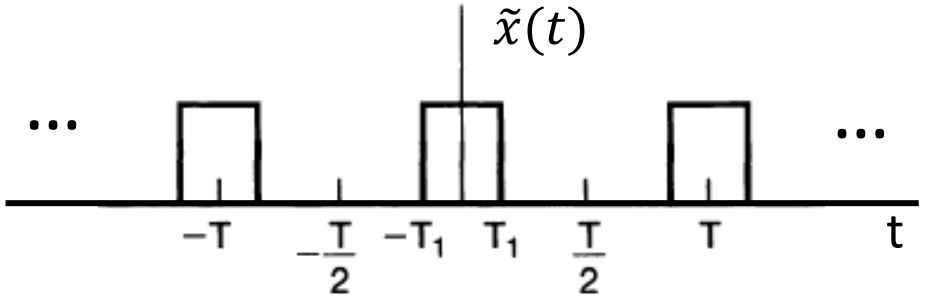
- Let's have a look at the inverse Fourier Transform of $2\pi\delta(\omega - k\omega_0)$:

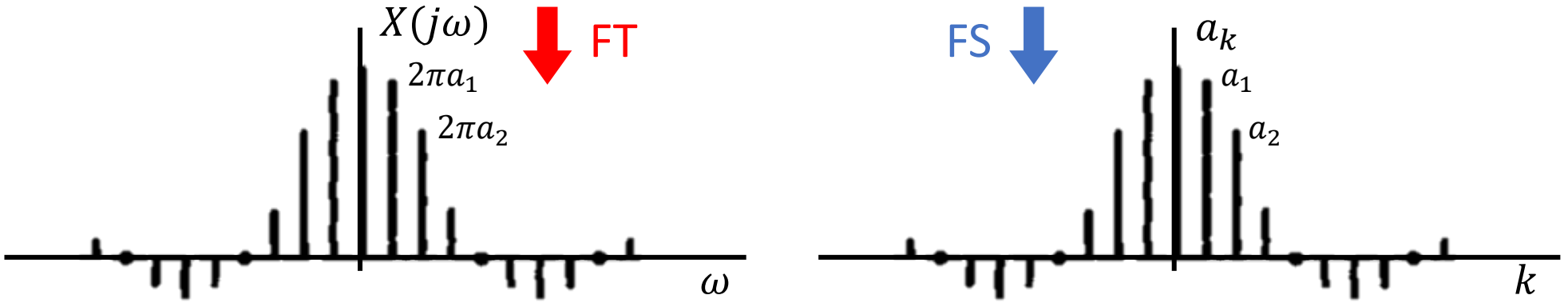
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\delta(\omega - k\omega_0) e^{j\omega t} d\omega = e^{jk\omega_0 t}$$

Fourier Transform vs. Fourier Series: The Relationship

- **For a periodic signal: Both Fourier Transform & Fourier Series are applicable**
  - Consider a periodic signal $x(t)$ with period T (thus $\omega_0 = 2\pi/T$):

$$\boldsymbol{X(j\omega) = \sum_{k=-\infty}^{+\infty} a_k 2\pi\delta(\omega - k\omega_0)}$$

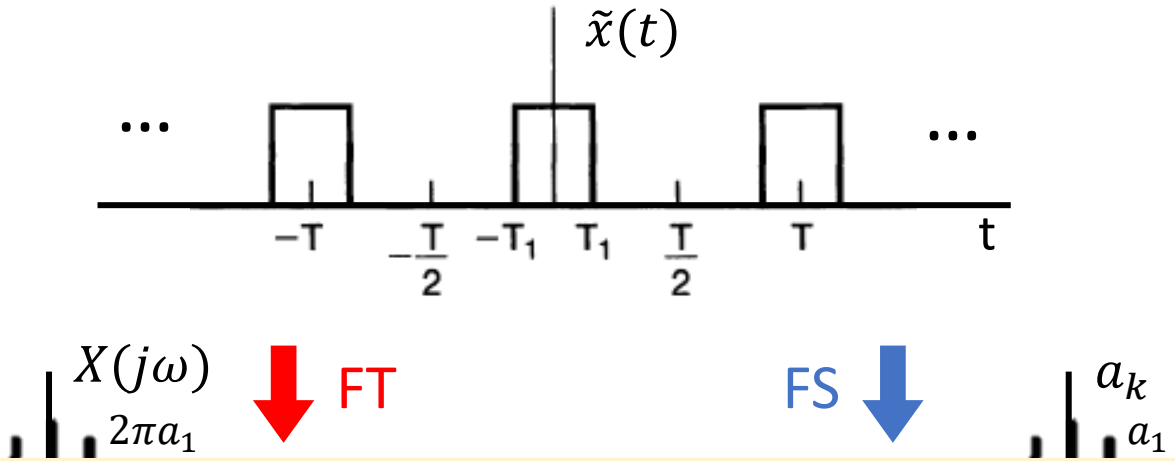




Fourier Transform vs. Fourier Series: The Relationship

- **For a periodic signal: Both Fourier Transform & Fourier Series are applicable**
  - Consider a periodic signal $x(t)$ with period T (thus $\omega_0 = 2\pi/T$):

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} a_k 2\pi\delta(\omega - k\omega_0)$$



Fourier Transform of a periodic signal with FS coefficients $\{a_k\}$ is **a train of impulses** occurring **at the harmonically related frequencies**, for which **the area of impulses at** the kth harmonic frequency $k\omega_0$ is $2\pi a_k$.

Fourier Transform: Examples

Q: Consider the signal:

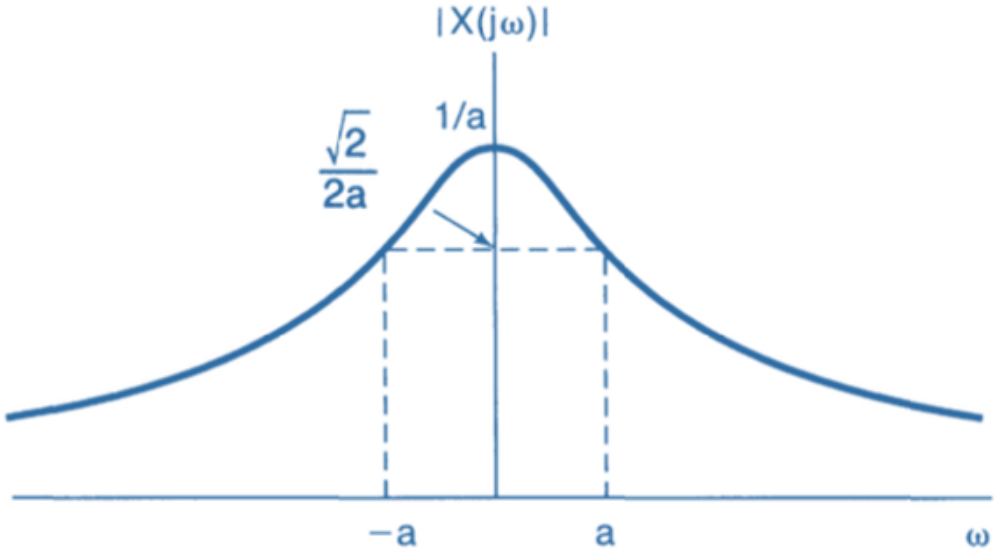
$$x(t) = e^{-at}u(t), \qquad a > 0$$

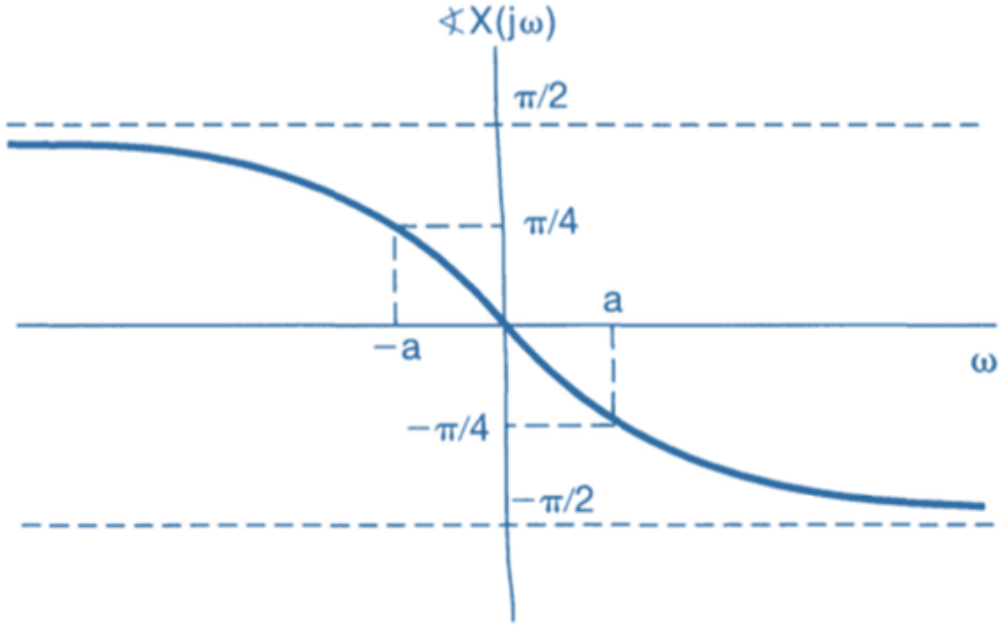
(Example 4.1)

Example 4.2

- Determine its Fourier Transform.

$$|X(j\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}, \quad \angle X(j\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$





Fourier Transform: Examples

Q: Determine the Fourier Transform of the unit impulse:

$$x(t) = \delta(t)$$

(Example 4.3)

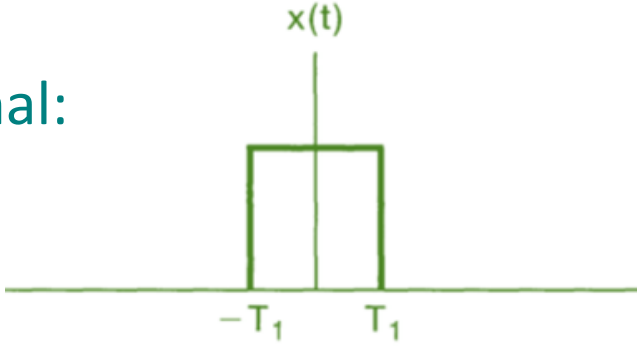
$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{+\infty} \delta(t)e^{-j\omega t}dt = 1$$

- The unit impulse has a Fourier transform consisting of equal contributions at *all* frequencies.

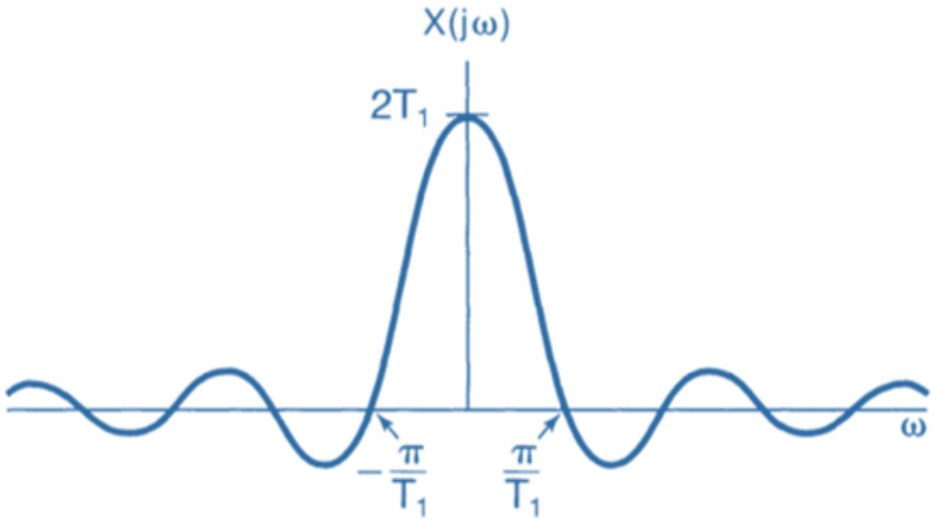
Fourier Transform: Examples

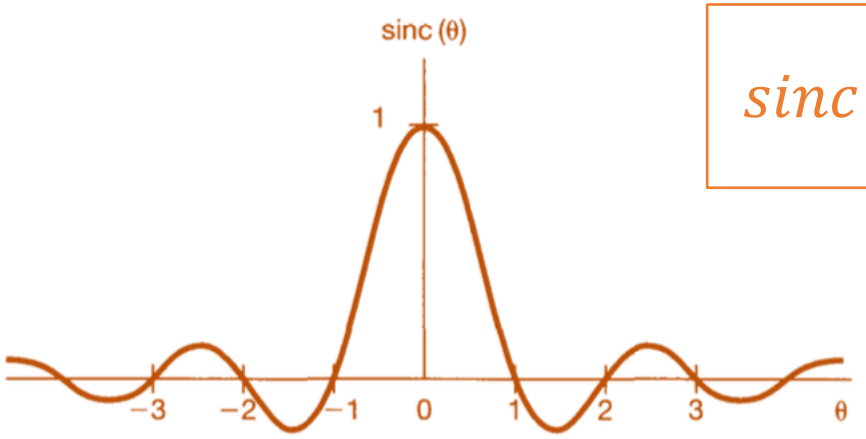
Q: Determine the Fourier Transform of the rectangular pulse signal:

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$



$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt = \int_{-T_1}^{+T_1} e^{-j\omega t}dt = \frac{2\sin(\omega T_1)}{\omega} = 2T_1 sinc(\frac{\omega T_1}{\pi})$$





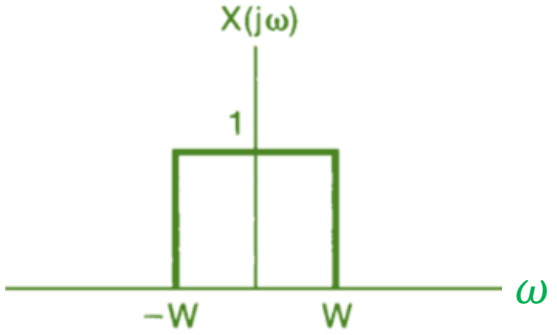
$$sinc(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$

(Example 4.4)

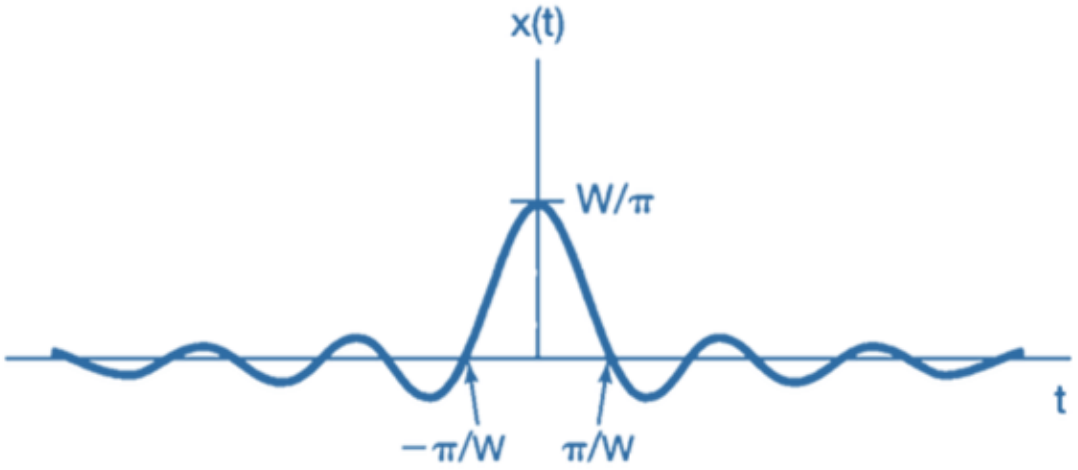
Fourier Transform: Examples

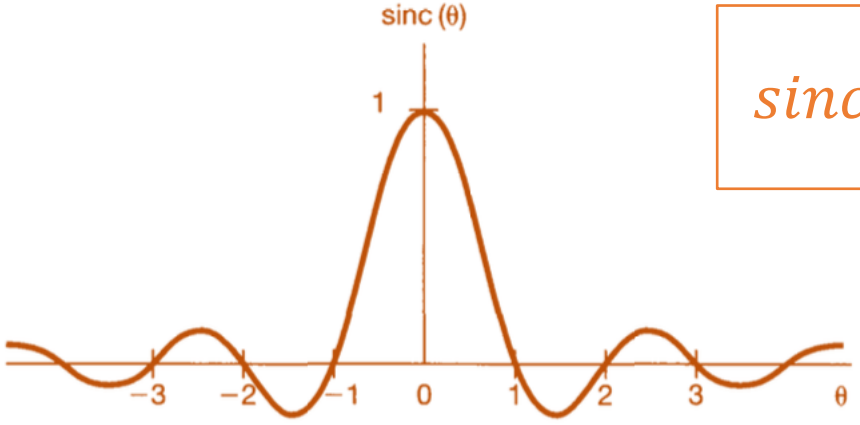
Q: Determine the signal $x(t)$ whose Fourier transform is:

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}d\omega = \frac{\sin(Wt)}{\pi t} = \frac{W}{\pi}sinc(\frac{Wt}{\pi})$$

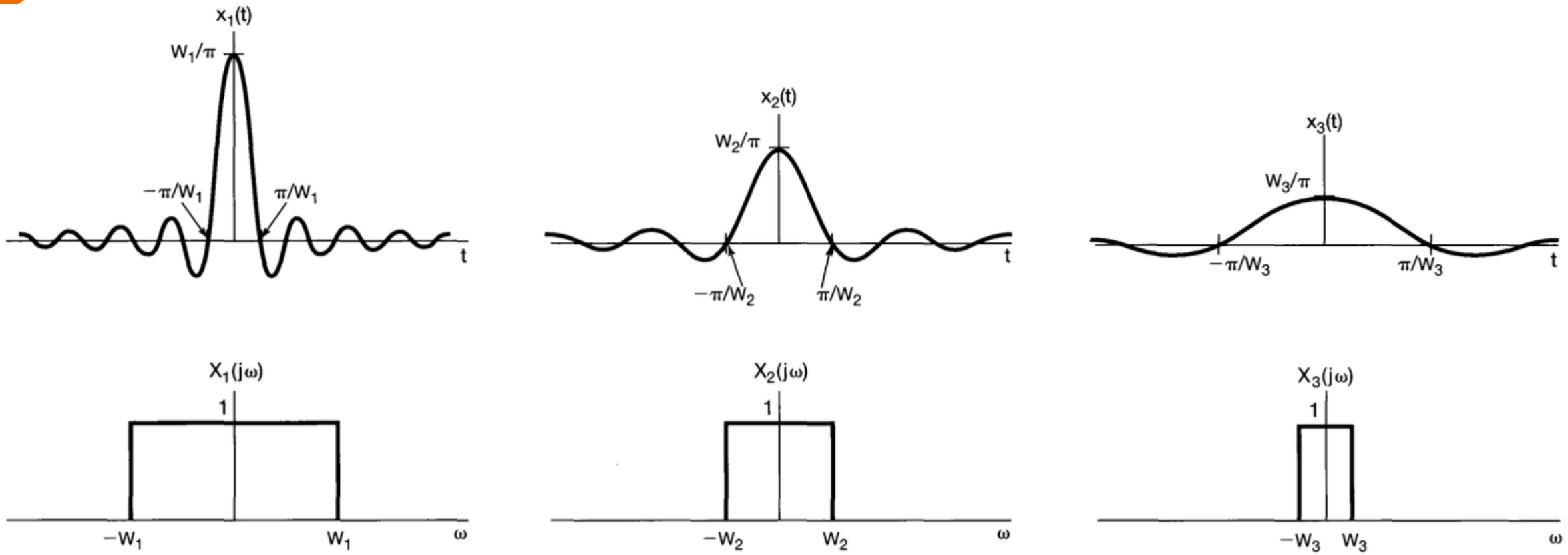




$$sinc(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$

(Example 4.4)

Fourier Transform: Examples



- As $W \uparrow$, $X(j\omega)$ becomes broader, while the main peak of $x(t)$ at $t = 0$ becomes higher and the width of the first lobe becomes narrower.
- In fact, as $W \to \infty$, $X(j\omega) = 1$ for all ω, $x(t)$ in Example 4.3 converges to an impulse.

FT of Periodic Signals: Examples

Q: Consider the signal:

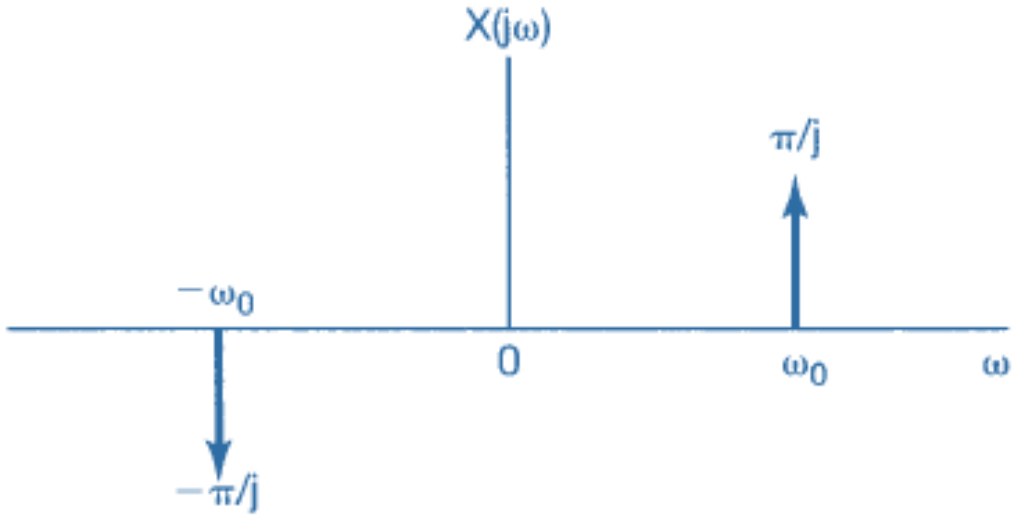
(Example 4.7)

$$x(t) = \sin(\omega_0 t)$$

- Determine its Fourier Transform.

Fourier coefficients: $a_1 = \frac{1}{2j}; \quad a_{-1} = -\frac{1}{2j}$

Fourier transform:



FT of Periodic Signals: Examples

Q: Determine the Fourier transform for the periodic signal below: (Example 4.8)

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Fourier coefficients:

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t - nT)\, dt = \frac{1}{T} \qquad a_k = \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t - nT) e^{-jk\omega_0 t}\, dt = \frac{1}{T}$$

Fourier transform:

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} a_k 2\pi\delta(\omega - k\omega_0) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{T})$$

- FT of periodic impulse train in the time domain → periodic impulse train in the frequency domain with period $2\pi/T$.
- Spacing between impulses in time domain ↑ → spacing in frequency domain ↓

Convergence of Fourier Transform

- **Dirichlet conditions** (*sufficient conditions*) under which a CT signal $x(t)$ is guaranteed to have a Fourier Transform:

1) $x(t)$ is absolutely integrable: $\int_{-\infty}^{\infty}|x(t)|dt < \infty$.
2) $x(t)$ has a finite number of maxima and minima in any finite interval.
3) $x(t)$ has a finite number of discontinuities in any finite interval, and each of these discontinuities is finite.

Q: How about the signal $x(t) = \frac{1}{t}u(t-1)$?

- **An alternative sufficient condition**: the signal have finite energy (*i.e., square integrable*): $\int_{-\infty}^{\infty}|x(t)|^2 dt < \infty$

Q: How about periodic signals?

→ A direct application of the FT equations to such signals will not necessarily yield a meaningful result → Relationship between FT and FS coefficients

Properties of CT Fourier Transform

- Recall:
  - Fourier Transform: $$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt$$
  - Inverse Fourier Transform: $$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega$$

- **Notation:**

$$\boldsymbol{x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)}$$

$$\boldsymbol{\mathcal{F}\{x(t)\} = X(j\omega)}$$

$$\boldsymbol{\mathcal{F}^{-1}\{X(j\omega)\} = x(t)}$$

Properties of CT Fourier Transform

- **Linearity**
- If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$$
$$y(t) \overset{\mathcal{F}}{\longleftrightarrow} Y(j\omega)$$

- Then

$$\boldsymbol{ax(t) + by(t) \overset{\mathcal{F}}{\longleftrightarrow} aX(j\omega) + bY(j\omega)}$$

- Proof?

Properties of CT Fourier Transform

- **Time Shifting**
- If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$$

- Then

$$\boldsymbol{x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_0} X(j\omega)}$$

- Proof?

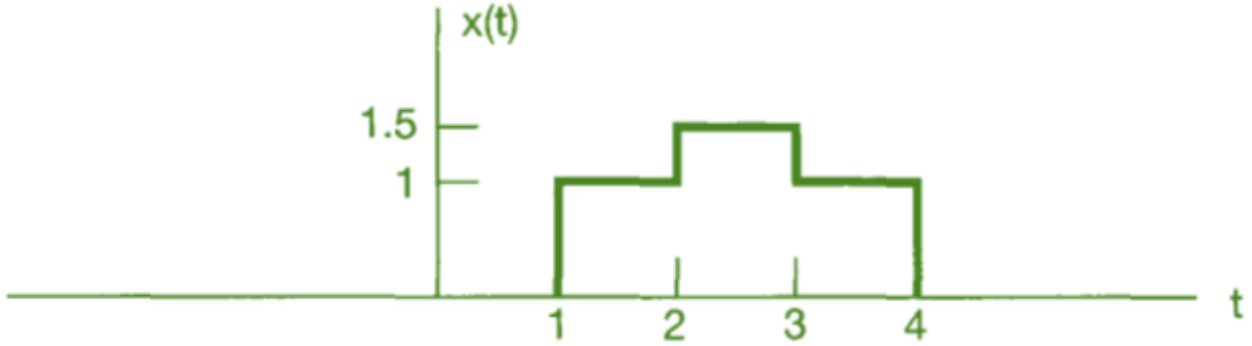
- Note that *a signal shifted in time* does not have *magnitude* of its FT altered.
- The effect of *a time shift* on a signal: to introduce a *phase shift into its FT* ($-\omega t_0$)

$$\boldsymbol{\mathcal{F}\{x(t-t_0)\} = e^{-j\omega t_0} X(j\omega) = |X(j\omega)| e^{j[\measuredangle X(j\omega) - \omega t_0]}}$$

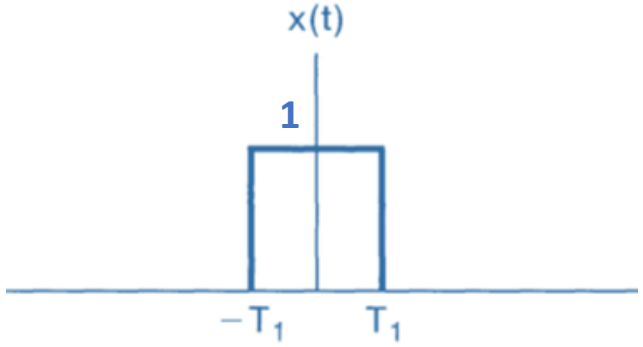
Properties of CT Fourier Transform

Q: Find the Fourier transform for the signal $x(t)$ below:

(Example 4.9)



- Recall the Fourier transform for a rectangular pulse signal



$$X(j\omega) = \frac{2\sin(\omega T_1)}{\omega}$$

- Apply the linearity and time shifting property?

Properties of CT Fourier Transform

- **Conjugation and Conjugate Symmetry**
- If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$$

- Then

Proof?

$$\boldsymbol{x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} X^*(-j\omega)}$$

- If $x(t)$ is real (*i.e.*, $x(t) = x^*(t)$), then $\boldsymbol{X(j\omega) = X^*(-j\omega)}$
  - Express: $X(j\omega) = Re\{X(j\omega)\} + jIm\{X(j\omega)\}$, then:
    $\boldsymbol{Re\{X(j\omega)\} = Re\{X(-j\omega)\}}$, and $\boldsymbol{Im\{X(j\omega)\} = -Im\{X(-j\omega)\}}$
    - Real part of the FT is an even function of frequency
    - Imag part of the FT is an odd function of frequency

Properties of CT Fourier Transform

- **Conjugation and Conjugate Symmetry**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
- Then

$$\boldsymbol{x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} X^*(-j\omega)}$$

Proof?

- If $x(t)$ is real (*i.e.*, $x(t) = x^*(t)$), then $\boldsymbol{X(j\omega) = X^*(-j\omega)}$
  - Express: $X(j\omega) = |X(j\omega)|e^{j[\measuredangle X(j\omega)]}$, then:
    - $|X(j\omega)|$ is an even function of frequency
    - $\measuredangle X(j\omega)$ is an odd function of frequency
  - Thus, for FT of a real signal, by specifying the real & imag (or mag & phase) of the transform for *positive frequencies*, those for *negative frequencies* can be derived directly.

Properties of CT Fourier Transform

- **Conjugation and Conjugate Symmetry**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
- Then

$$\boldsymbol{x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} X^*(-j\omega)}$$

Proof?

- If $x(t)$ is real (*i.e.*, $x(t) = x^*(t)$), then $\boldsymbol{X(j\omega) = X^*(-j\omega)}$
  - If $x(t)$ is real and even, $X(j\omega)$ is real and even
  - If $x(t)$ is real and odd, $X(j\omega)$ is purely imaginary and odd
  - Express $x(t)$ in terms of the sum of an even function and an odd function
    - $\boldsymbol{Ev\{x(t)\} \overset{\mathcal{F}}{\longleftrightarrow} Re\{X(j\omega)\}}$
    - $\boldsymbol{Od\{x(t)\} \overset{\mathcal{F}}{\longleftrightarrow} jIm\{X(j\omega)\}}$

Properties of CT Fourier Transform

- **Differentiation and Integration**
- If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$$

- Then

$$\frac{dx(t)}{dt} \overset{\mathcal{F}}{\longleftrightarrow} j\omega X(j\omega)$$

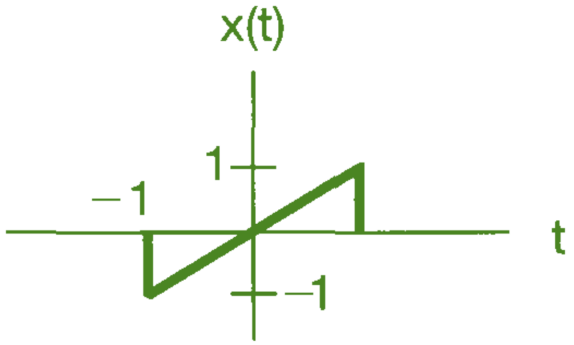
$$\int_{-\infty}^{t} x(\tau)d\tau \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$$

- Transforms the operations of differentiation/integration in time domain into multiplication/division by $j\omega$ in the frequency domain.

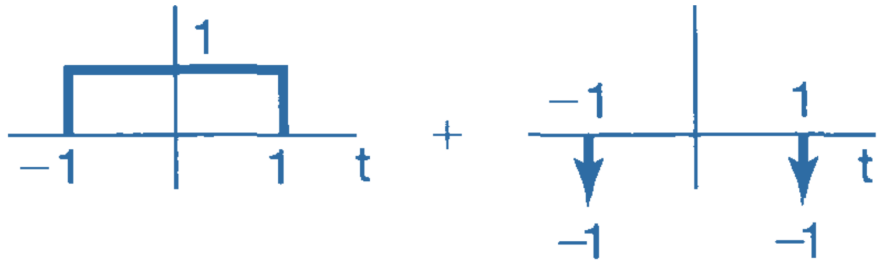
Properties of CT Fourier Transform

Q: Find the Fourier transform for the signal $x(t)$ below:

Example 4.11
(Example 4.12)



- Let's use the differentiation & integration property: $g(t) = \frac{dx(t)}{dt}$



- Is $X(j\omega)$ even or odd? Is it real or purely imaginary?

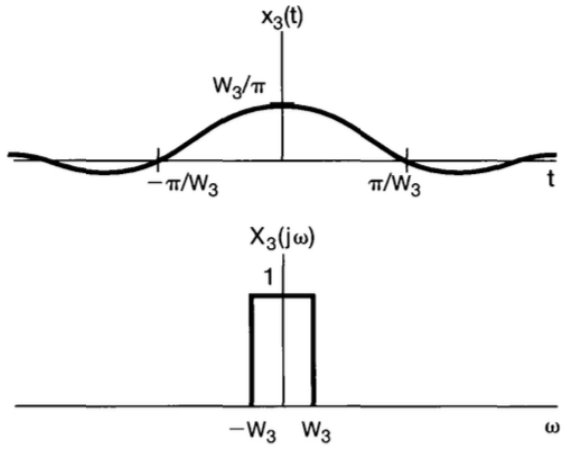
Properties of CT Fourier Transform

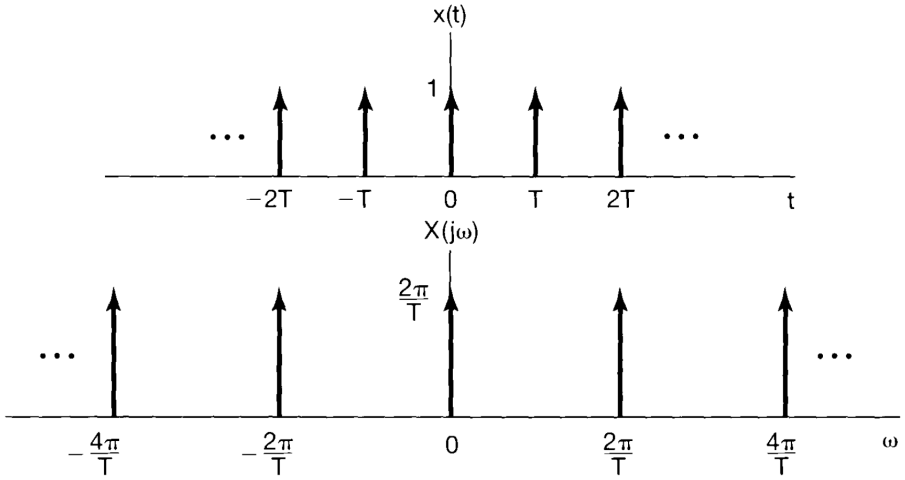
- **Time and Frequency Scaling**

- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
- Then

$$\boldsymbol{x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)} \quad (a \text{ is a non-zero real number})$$

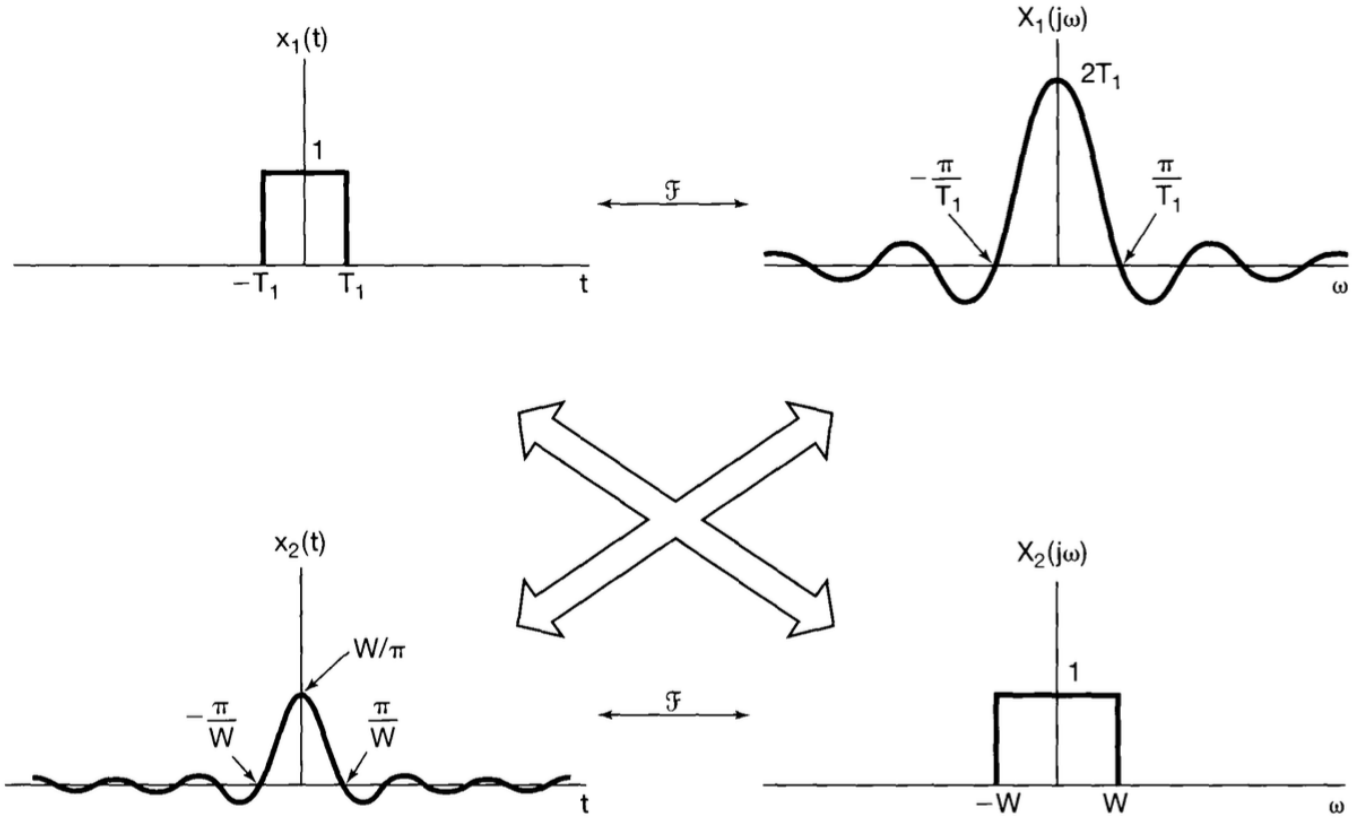
- With $a = -1$, $\boldsymbol{x(-t) \overset{\mathcal{F}}{\longleftrightarrow} X(-j\omega)}$





Properties of CT Fourier Transform

- **Duality**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
- Then $\boldsymbol{X(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x(-\omega)}$



Properties of CT Fourier Transform

- **Duality**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$, then $\boldsymbol{X(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x(-\omega)}$
- Proof:
- By synthesis equation:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega \Longrightarrow 2\pi x(t) = \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega$$

- Replace t with $-t$:

$$2\pi x(-t) = \int_{-\infty}^{+\infty} X(j\omega)e^{-j\omega t}d\omega$$

- Interchange the variables t and ω:

$$2\pi x(-\omega) = \int_{-\infty}^{+\infty} X(jt)e^{-j\omega t}dt$$

Properties of CT Fourier Transform

Q: Given $x(t) = e^{-|t|} \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega) = \frac{2}{1+\omega^2}$

Use duality to find the Fourier transform of $g(t) = \frac{2}{1+t^2}$

- By synthesis equation:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{2}{1+\omega^2} e^{j\omega t} d\omega \Longrightarrow 2\pi x(t) = \int_{-\infty}^{+\infty} \frac{2}{1+\omega^2} e^{j\omega t} d\omega$$

- Replace t with $-t$:

$$2\pi x(-t) = \int_{-\infty}^{+\infty} \frac{2}{1+\omega^2} e^{-j\omega t} d\omega$$

- Interchange the variables t and ω:

$$2\pi x(-\omega) = \int_{-\infty}^{+\infty} \frac{2}{1+t^2} e^{-j\omega t} dt = 2\pi e^{-|\omega|}$$

Properties of CT Fourier Transform

- **Duality** also *applies* to properties of the Fourier transform
- For example:

$$x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_0}X(j\omega)$$

$$e^{j\omega_0 t}x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j(\omega-\omega_0))$$

$$\frac{dx(t)}{dt} \overset{\mathcal{F}}{\longleftrightarrow} j\omega X(j\omega)$$

$$-jtx(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{dX(j\omega)}{d\omega}$$

$$\int_{-\infty}^{t} x(\tau)d\tau \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{j\omega}X(j\omega)+\pi X(0)\delta(\omega)$$

$$-\frac{1}{jt}x(t)+\pi x(0)\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} \int_{-\infty}^{\omega} X(\eta)d\eta$$

Properties of CT Fourier Transform

- **Parseval's Relation**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
- Then

$$\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$$

- Proof?

- Total energy could be determined by:

1) (LHS): Integrate energy per unit time ($|x(t)|^2$) over all time
2) (RHS): Integrate energy per unit frequency ($|X(j\omega)|^2/2\pi$) over all frequencies

$|X(j\omega)|^2$ also referred to as *energy-density spectrum* of the signal $x(t)$.

Properties of CT Fourier Transform

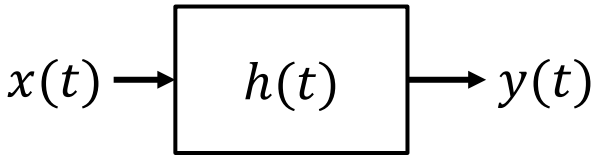
- **The Convolution Property**
- If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega), \quad h(t) \overset{\mathcal{F}}{\longleftrightarrow} H(j\omega)$
- Then

$$\boldsymbol{y(t) = h(t) * x(t) \overset{\mathcal{F}}{\longleftrightarrow} Y(j\omega) = H(j\omega)X(j\omega)}$$

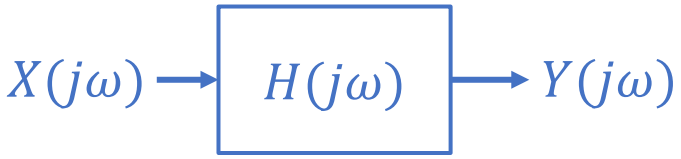
- Proof?

Properties of CT Fourier Transform

- **The Convolution Property**
- Recall for LTI systems:



$$y(t) = x(t) * h(t)$$



$$Y(j\omega) = X(j\omega)H(j\omega) \qquad H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

- where the frequency response $H(j\omega)$:

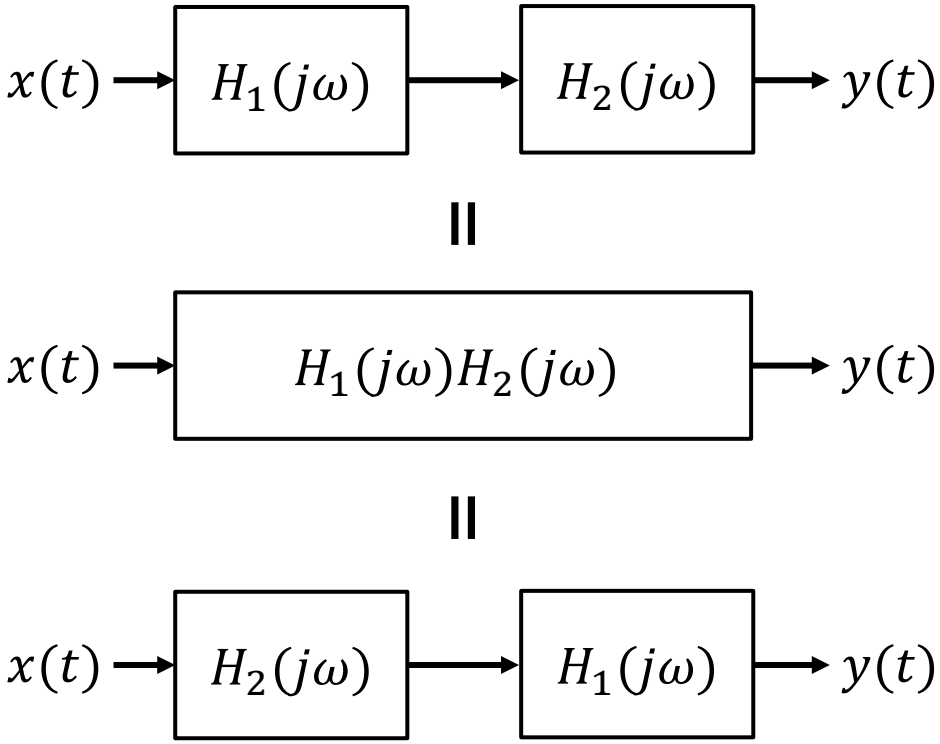
$$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)e^{-j\omega\tau}d\tau$$

  - Assuming the LTI system is stable & satisfies the 2 other Dirichlet conditions:

$$\int_{-\infty}^{+\infty} |h(t)|dt < \infty \quad \text{(LTI system stability)}$$

Properties of CT Fourier Transform

- **The Convolution Property**
- For LTI systems:



Properties of CT Fourier Transform

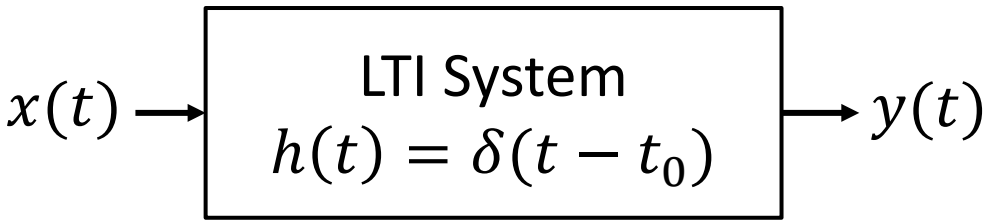
Q: Consider a CT LTI system with impulse response

$$h(t) = \delta(t - t_0)$$

(Example 4.15)

Given an input $x(t)$, find the output $y(t)$ of this system.

- By convolution?
- By properties of the Fourier transform?



$$\boldsymbol{y(t) = x(t - t_0)}$$

Properties of CT Fourier Transform

Q: Consider a CT LTI system with impulse response

$$h(t) = e^{-at}u(t), \qquad a > 0$$

(Example 4.19)

Find the system output $y(t)$ to the input signal

$$x(t) = e^{-bt}u(t), \qquad b > 0$$

- By properties of the FT
- Techniques of Partial fraction expansion

$$y(t) = \frac{1}{b-a}[e^{-at}u(t) - e^{-bt}u(t)] \qquad a \neq b$$

$$y(t) = te^{-at}u(t) \qquad a = b$$

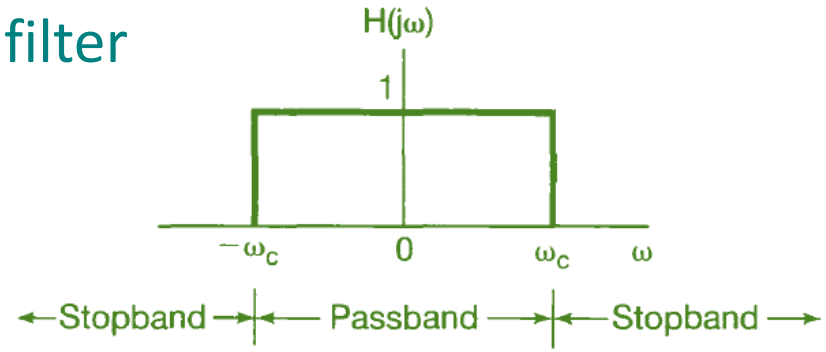
Properties of CT Fourier Transform

Q: Determine the response of an ideal lowpass filter to an input signal $x(t)$:

$$x(t) = \frac{\sin(\omega_i t)}{\pi t}$$

(Example 4.20)

- Ideal low-pass filter



$$h(t) = \frac{\sin(\omega_c t)}{\pi t}$$

- Use the convolution property

$$y(t) = \begin{cases} \dfrac{\sin \omega_c t}{\pi t} & \text{if } \omega_c \leq \omega_i \\ \dfrac{\sin \omega_i t}{\pi t} & \text{if } \omega_i \leq \omega_c \end{cases}$$

Properties of CT Fourier Transform

- **The Multiplication Property**
- If $x_1(t) \overset{\mathcal{F}}{\longleftrightarrow} X_1(j\omega), \quad x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} X_2(j\omega)$
- Then

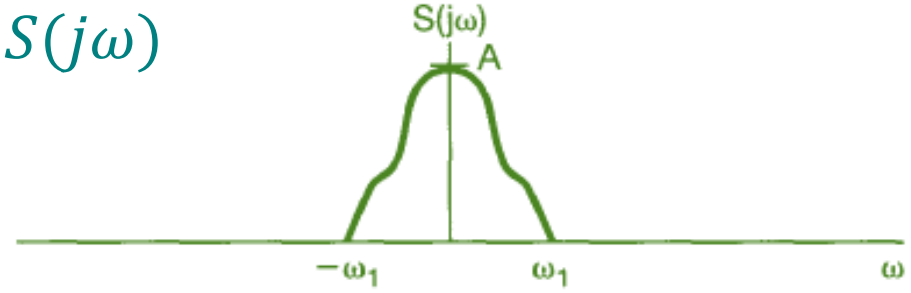
$$\boldsymbol{y(t) = x_1(t)x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} Y(j\omega) = \frac{1}{2\pi}[X_1(j\omega) * X_2(j\omega)]}$$

$$\boldsymbol{= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(j\theta)X_2(j(\omega-\theta))d\theta}$$

- The multiplication of two signals: often *referred to* as amplitude modulation
- Thus, this property is also *referred* to as the modulation property

Properties of CT Fourier Transform

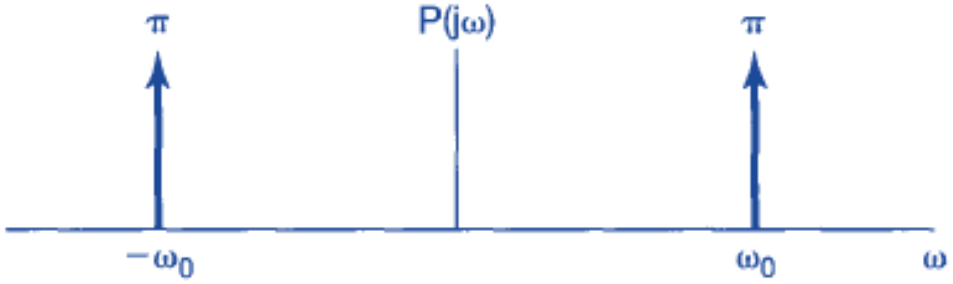
Q: Given $s(t)$ with the spectrum $S(j\omega)$

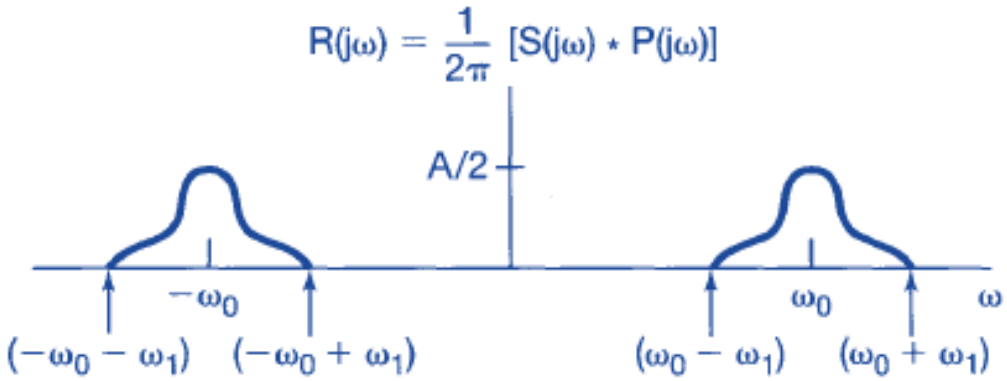


(Example 4.21)

Consider the signal $p(t) = cos(\omega_0 t)$.

Sketch the spectrum: $R(j\omega)$ of $r(t) = s(t)p(t)$.





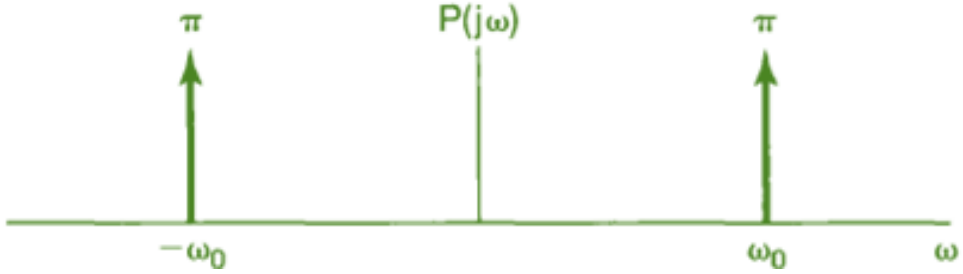
Properties of CT Fourier Transform

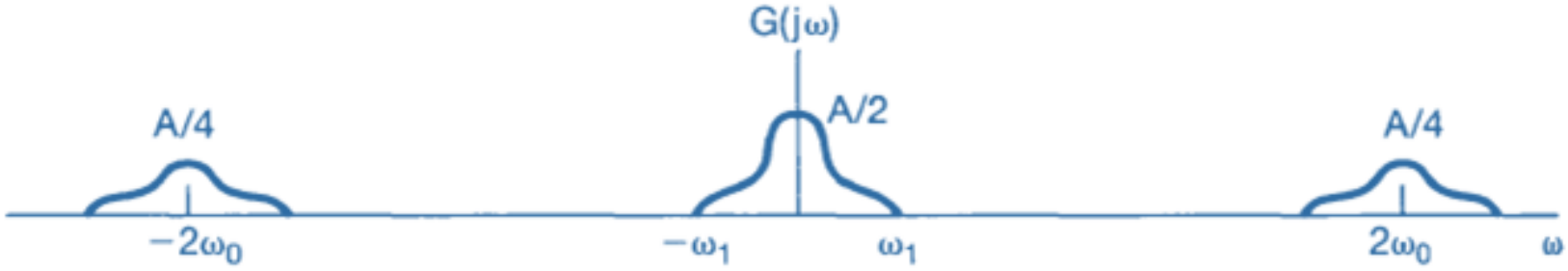
Q: Given $s(t)$ with the spectrum $S(j\omega)$, and signal $p(t) = cos(\omega_0 t)$.

Sketch the spectrum: $R(j\omega)$ of $r(t) = s(t)p(t)$.

(Example 4.22)







Properties of CT Fourier Transform

- **Summary (Read Table 4.1, Table 4.2 in the textbook!)**

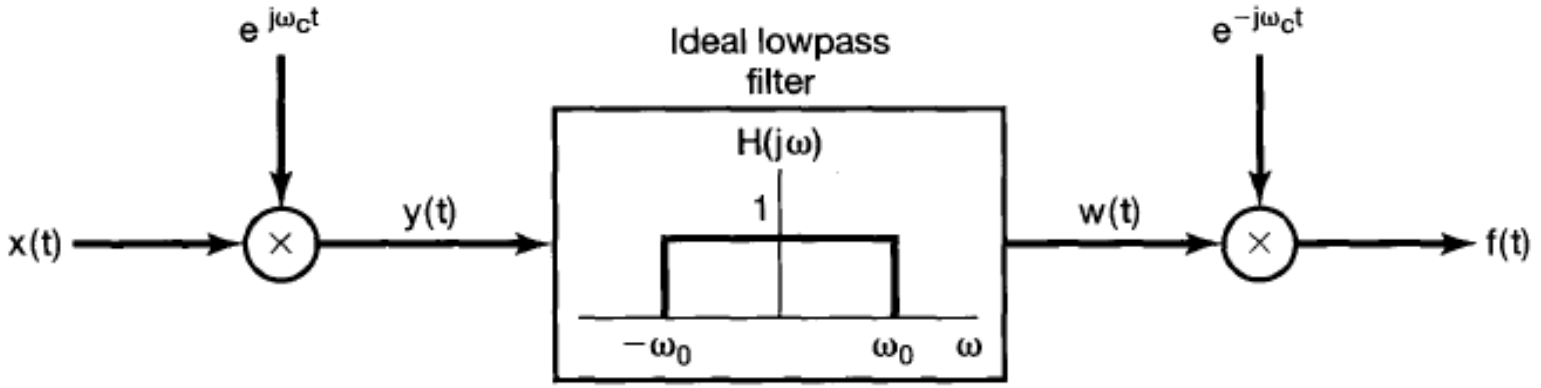
TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$	$X(j\omega)$
		$y(t)$	$Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\theta)Y(j(\omega-\theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^{t}x(t)dt$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$X(j\omega) = X^*(-j\omega)$ $\mathcal{R}e\{X(j\omega)\} = \mathcal{R}e\{X(-j\omega)\}$ $\mathcal{I}m\{X(j\omega)\} = -\mathcal{I}m\{X(-j\omega)\}$ $\lvert X(j\omega)\rvert = \lvert X(-j\omega)\rvert$ $\measuredangle X(j\omega) = -\measuredangle X(-j\omega)$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\mathcal{R}e\{X(j\omega)\}$ $j\mathcal{I}m\{X(j\omega)\}$

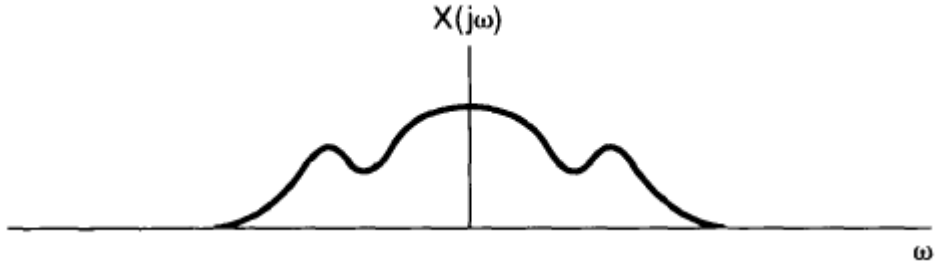
TABLE 4.2 BASIC FOURIER TRANSFORM PAIRS

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty}a_k e^{jk\omega_0 t}$	$2\pi\sum_{k=-\infty}^{+\infty}a_k\delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos\omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin\omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi\delta(\omega)$	$a_0 = 1, \quad a_k = 0, \quad k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave $x(t) = \begin{cases} 1, & \lvert t\rvert < T_1 \\ 0, & T_1 < \lvert t\rvert \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty}\frac{2\sin k\omega_0 T_1}{k}\delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi}\operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty}\delta(t - nT)$	$\frac{2\pi}{T}\sum_{k=-\infty}^{+\infty}\delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t)\begin{cases} 1, & \lvert t\rvert < T_1 \\ 0, & \lvert t\rvert > T_1 \end{cases}$	$\frac{2\sin\omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \lvert\omega\rvert < W \\ 0, & \lvert\omega\rvert > W \end{cases}$	—

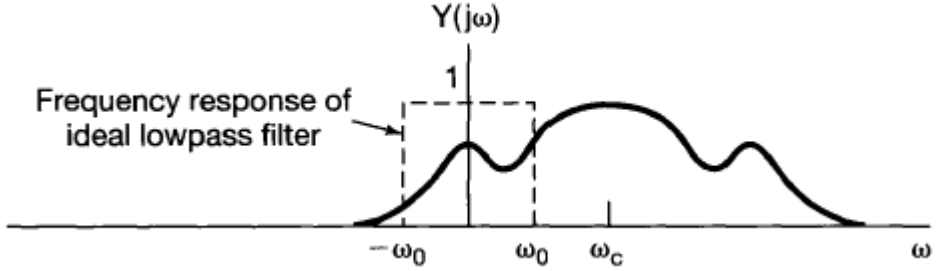
Frequency-Selective Filtering with Variable Center Frequency



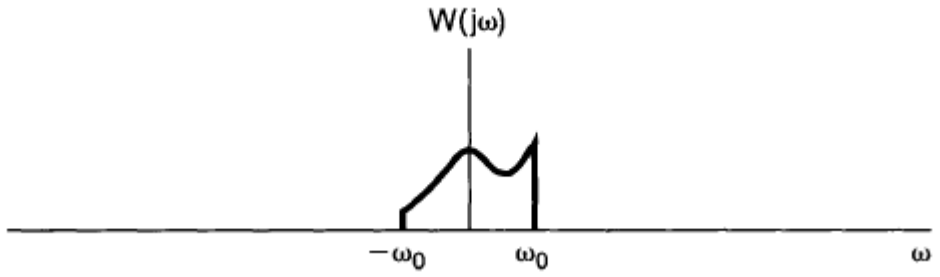
- Given the spectra $X(j\omega)$ of the signal $x(t)$:



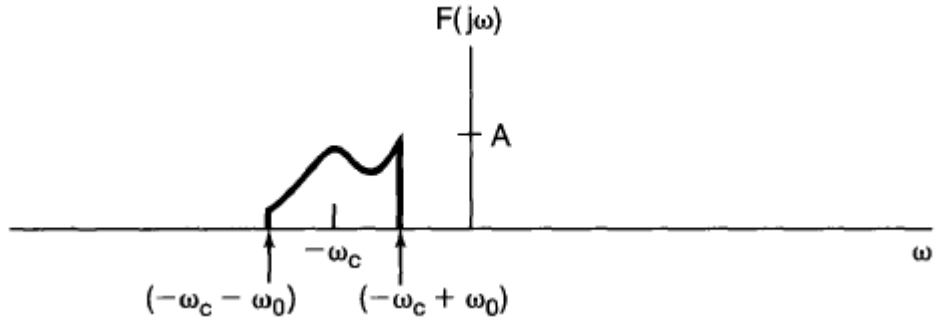
$y(t)$



$w(t)$



$f(t)$



Summary

- **Continuous-Time Fourier Transform**
  - Fourier transform and inverse Fourier transform for aperiodic signals
  - Fourier transform for periodic signals, relationship with Fourier coefficients

- **Properties of CT Fourier Transform**
  - Linearity, time & frequency shifting, conjugation, time & frequency scaling, convolution, multiplication, differentiation & integration, duality …